Quantum Gravity in Fewer Effective Dimensions: What the Reductions Explain

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Abstract

A space can have four coordinates yet behave as if it has fewer dimensions in a particular small-scale measurement. Could that help reconcile gravity with quantum physics? A worked example explains this effective reduction and distinguishes it from removing coordinate directions. The paper also examines a result about unitarity—the consistency of quantum probabilities—order by order in a specified approximation. It then compares seven approaches by the physics they recover and the assumptions they still need. Mathematical monism places gravity and quantum theory within one reality; a successful unification must supply the common dynamics. The contribution is a clear comparison of what dimensional reduction establishes and what a complete theory must add.

Keywords: mathematical monism; spectral dimension; quantum gravity; unitarity; theory selection

1. One ontology, several dynamics

If reality is one mathematical structure, gravity and quantum physics must belong to the same account. The remaining task is to find their common rules. One possibility studied here is that the effective number of dimensions changes with scale. The paper separates what that idea can explain from what existing theorems establish.

It does not identify two equations merely because both are mathematical. The equations x-dot = x and x-dot = 2x have the same kind of mathematical variables and different solutions. A shared ontology leaves room for incompatible models. This elementary distinction explains why the metaphysical unification can be argued a priori while a derivation connecting general relativity and quantum theory still matters.

The requirement is not that a mathematical proof must be verified by experiment. It is that a conclusion be derived from its actual premises. Deriving an observed limit from a proposed action is itself mathematical work. Comparing the resulting predictions with observations addresses a different question: whether this is the structure occupied by us.

2. What is becoming lower dimensional?

Carlip's reviews distinguish several notions of dimension and collect evidence for short-distance dimensional reduction in multiple quantum-gravity programmes.

These are related clues, not seven independent measurements of one fundamental line. Coordinate dimension, spectral dimension, the dimension of a graph edge and the dimension of a string worldsheet answer different questions.

2.1 A worked example

Consider a positive Euclidean diffusion generator with momentum dependence |k|^(2z), where z > 0, on d coordinate dimensions. Its return probability per unit volume has the form

$$P(s) = C \int_{\mathbb{R}^d} d^d k \, e^{-s|k|^{2z}}.$$

Rescale k = s^(-1/(2z))q. For s > 0 the radial integral converges and

$$P(s) = C' s^{-d/(2z)}.$$

The spectral dimension is therefore

$$D_s = -2 \frac{d \log P}{d \log s} = \frac{d}{z}.$$

With d = 4 and z = 2 it equals two although the integration still uses four coordinates. A kinetic operator that crosses from $|k|^2$ at low momentum to |k| at high momentum can accordingly have different effective spectral dimensions at different diffusion scales.

This calculation is a diagnostic example, not a Lorentzian quantum-gravity theory. In particular it proves neither scattering unitarity nor the absence of unwanted modes. Its value is to isolate an inference: a measured or calculated spectral dimension of two does not, by itself, imply that physical objects are made of two-dimensional pieces, or of one-dimensional numbers.

2.2 The approaches actually listed in this portfolio

Approach	Relevant structure	What must be distinguished
String theory	One-dimensional strings sweeping two-dimensional worldsheets	String and worldsheet dimensions versus spacetime and spectral dimensions
Loop quantum gravity	Spin networks and geometric operators	Graph edges, area and volume spectra, and recovered spacetime geometry

Approach	Relevant structure	What must be distinguished
Causal dynamical triangulations	A sum over triangulated geometries	The lattice regulator and the proposed scaling limit
Asymptotic safety	Renormalization-group behaviour	A fixed-point proposal and an effective dimension inferred from it
Hořava–Lifshitz gravity	Anisotropic scaling of space and time	Scaling exponents versus the number of coordinates
Causal sets	A locally finite causal order	Discrete order, volume counting and manifold approximation
Fractional field theory	Fractional kinetic operators	Operator scaling, physical spectrum and amplitude prescription

This comparison does not claim that all seven approaches satisfy every premise proposed by Matheism. The listed dimensions and consistency results concern different mathematical structures and must be assessed on their own terms.

3. The recent theorem

Calcagni and Briscese (2026) study specified fractional scalar, gauge and gravitational theories. Their perturbative unitarity results depend on the kinetic operator, exponent range and amplitude prescription. The hermitian construction uses the fakeon prescription; the non-hermitian case has stated ranges compatible with the Feynman prescription. The article is published as JHEP09(2026)206.

This establishes a consistency result within the analysed class. It neither supplies a result for every programme in the table nor derives Standard Model parameters. The fakeon literature also discusses violations of microcausality; calling that prescription microcausality-preserving reverses a stated feature of the approach (Anselmi 2019).

It is a substantial gain to have a consistent candidate class. A programme can now investigate that class without treating its previously unresolved unitarity issue as an automatic reason for dismissal. Questions of ultraviolet behaviour, physical limits and phenomenology must still be answered for the chosen model and prescription.

4. Numbers specify particles; a parameter list is not a particle theory

The Standard Model is more than its numerical parameters. It specifies fields, transformation laws, interactions and a state space. A coupling or mass is a scalar parameter. An electron is not a scalar field, and a gauge boson does not become a scalar particle because its properties can be encoded by numbers. The distinction is mathematical throughout; it introduces no extra substance.

The strongest monist statement is consequently that the complete mathematical specification constitutes the physical content. It cannot be compressed to a parameter list by omitting the structures that give the parameters their meaning. Nor does a small observed particle spectrum prove that no simpler generating rule exists beneath the effective description. Such a rule could constrain its outputs strongly.

5. What can be excluded a priori?

An explicit premise can exclude a candidate from a research programme. A contradiction in the candidate's formalism can exclude that formulation. These are different exclusions.

If fundamental physical discreteness is imposed, a fundamental continuum ontology is outside the programme. A continuum approximation remains available. A lattice theory that takes a continuum scaling limit may therefore conflict with the premise at its proposed fundamental endpoint while still supplying useful effective calculations. Loll (2020) describes precisely such a scaling-limit aim for causal dynamical triangulations. Whether a discrete alternative can recover the same successful limit becomes a constructive question.

The argument that zero-measure points cannot yield positive length does not itself prove the premise. Countable additivity does not extend to arbitrary uncountable unions. Using that requirement as a selection principle is stronger than attributing a contradiction to measure theory that it does not contain.

Rejecting fundamental stochastic outcomes rules out collapse laws with those outcomes within the proposed programme. It does not select superdeterminism by logic alone. Bell's statistical independence condition concerns the joint distribution of hidden variables and settings, not whether either is produced by a deterministic algorithm. A deterministic common generator can produce dependent or independent outputs.

Everett's unitary branching and a local single-outcome hidden-variable model relax different assumptions in familiar Bell arguments. They can be compared without treating their conjunction as already forced by the block universe. Likewise a short rule for digits of pi establishes computability, not normality, cryptographic pseudorandomness or a distribution matching all quantum experiments.

6. Parsimony with complete candidates

Fix a prefix description language U and specify candidates to the same observational precision. Under a shortest-description prior proportional to 2^(-K_U), an established difference of Δ bits gives an individual prior ratio of 2^{Δ} in favour of the shorter candidate. A comparison using supplied programs instead of minimum lengths is a comparison of those descriptions. Both comparisons need a declared decoder, data convention and scope.

Counting 10^500 alternatives gives a fixed-length indexing cost of 1,661 bits. It does not prove that the particular alternative occupied by us needs such an index: a short rule could select it. Conversely, a short list of constants is not a complete competing account if interactions, geometry or boundary data have been left outside the charge.

The positive programme is to exploit shared structure, derive as much as possible, and count the residual choices honestly. A candidate that derives the same observations with fewer independent specifications earns a real advantage. Mathematical monism motivates this search without presupposing its winner.

7. Conclusion

A theory can behave as if it has fewer dimensions without losing coordinate directions. The diffusion example makes that distinction explicit. The probability-consistency theorem examined here establishes a further result for its specified class of models.

Together these results sharpen the search for a common account of gravity and quantum physics. To complete it, a candidate must derive the relevant particles and interactions, recover the established limits and explain the observations. Mathematical monism gives that search one proposed reality to explain.

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